GOKHAN SAHIN

Manufacturing Engineer · Production Planning · ERP Systems Developer
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Authorized to work in the United States · Open to relocation

PROFESSIONAL SUMMARY

Manufacturing Engineer with a production planning foundation, combining hands-on process improvement on the shop floor with advanced ERP systems development. Experienced in root-cause analysis, BOM architecture, process workflow design, and lean implementation (Kanban, 5S, Kaizen) across aerospace (AS9100) and Tier 1 automotive manufacturing environments. Led an enterprise ERP migration, built and deployed custom AL extensions in Microsoft Dynamics 365 Business Central, and architected an AI-driven process automation platform. Six Sigma Green Belt with experience managing 100+ production personnel across multi-shift operations in the U.S., Mexico, and Turkey. ABET-accredited Management Engineering degree (Industrial Engineering, Operations Research, Manufacturing Systems). Trilingual: English, Spanish, Turkish.

CORE COMPETENCIES

Manufacturing Process Design & Improvement · Root Cause Analysis · BOM Management · Process Workflow Design
Production Scheduling & Capacity Planning · Lean Manufacturing (5S, Kaizen, Kanban) · Six Sigma Green Belt
MRP/ERP Systems (Business Central, AX, SAP) · AL Extension Development · Python, SQL, PowerBI, MATLAB
AS9100/ISO 9001 Compliance · Inventory Control · Subcontracting & Outsource Process Management · FMEA Awareness

PROFESSIONAL EXPERIENCE

Manufacturing / Production Planning Engineer — Apr 2025 – Present

Air Electro, Inc. – Los Angeles, CA (AS9100/ISO 9001 Aerospace Manufacturer)

- **Own end-to-end manufacturing process for 900+ active part numbers across 8+ work centers,** including fabrication, assembly, and subcontracted operations supporting F-35, Apache AH-64, 787, and other defense/commercial programs.
- **Designed and implemented multi-level BOM architectures** for complex assemblies with up to five cascading production order levels, including subcontracted plating operations with scrap tracking isolated as a separate P&L line item for accurate cost analysis.
- **Led the company's ERP migration from AX Dynamics 365 to Microsoft Business Central,** re-engineering the order fulfillment workflow end-to-end, training the sales team, and resolving post-migration issues across warehouse shipments, item tracking, and lot/bin reconciliation.
- **Built and deployed custom AL extensions to production** via VS Code, including a Split Production Order tool with reservation continuity and on-hand inventory FlowFields, eliminating manual inventory lookups and reducing production order processing time.
- **Executed a bin-by-bin inventory reconciliation across ~21,000 lines,** performing root-cause analysis on AX migration discrepancies and implementing a two-phase quarantine-and-reload correction strategy that restored inventory accuracy.
- Designed custom report layouts (RDLC) and automated Excel/PowerQuery dashboards for production KPIs, backlog aging, and on-time delivery, adopted as the primary reporting tool across all departments.
- Developed a three-state demand classification model (Physical Reserved, Reserved Ordered, On Order) improving inventory allocation accuracy and reducing over-commitment of available stock for production.

Production Planning Engineer / Process Improvement Lead — Sep 2023 – Jun 2024

Teklas Automotive (Tier 1 OEM Supplier to VW, Ford, Stellantis) – Aguascalientes, Mexico

- **Reduced scheduling errors by ~30%** through root-cause analysis of ERP data misalignment, standardizing planning procedures and production routing across extrusion and vulcanization process lines.
- **Optimized inventory levels by ~18%** by designing and implementing Kanban pull systems on the shop floor, eliminating overstock while maintaining zero production stoppages due to material shortages.
- **Established a rapid-response bottleneck resolution protocol** achieving sub-24-hour resolution time for production disruptions, improving overall equipment uptime and reducing unplanned downtime across all shifts.
- **Managed 100+ production personnel** across multi-shift extrusion and vulcanization operations, coordinating daily production scheduling, shift handoffs, and cross-functional communication between maintenance, quality, and logistics teams.
- Designed custom ERP interfaces (SQL-based) and interactive capacity planning dashboards in Excel/PowerBI, becoming the primary daily tools for plant leadership and enabling data-driven manufacturing decisions.

Demand Planning & Supply Chain Analyst — Jan 2023 – Sep 2023

Mondelez International (Fortune 500 FMCG) – Istanbul, Turkey

- Analyzed promotional demand data and sell-through patterns to generate demand forecasts, optimizing production-to-shelf availability and reducing out-of-stock incidents for high-velocity SKUs.
- Coordinated order fulfillment workflows between manufacturing plants, warehousing, and distribution logistics, improving on-time delivery across a multi-market network.
- Served as operational link between supply chain planning and commercial teams, managing key account escalations within SLA targets for a portfolio generating $1B+ in regional revenue.

TECHNICAL PROJECTS

CENTA – AI-Driven Manufacturing Process Automation | *Python, Pandas, BC Standard API v2.0, OData, LLaMA, OCR*

- **Architected a modular Python automation pipeline** integrating with Microsoft Business Central's Standard API v2.0 to automate repetitive manufacturing administration tasks in an AS9100 environment.
- Built Sales Quote auto-generation from customer emails and a read-only Sales Order lifecycle tracker merging data from six BC entities via OData, reducing quote turnaround from hours to minutes.
- Designed for future migration from OData to custom AL API pages ahead of the 2027 OData deprecation timeline.

Custom ERP Pages & Manufacturing Dashboards | *AX Dynamics 365, SQL-Based ERP, PowerBI, Excel/PowerQuery*

- Designed and built role-specific ERP interfaces at Teklas (SQL-based) and Air Electro (AX/BC) that replaced manual lookups and became primary daily tools for production, sales, and plant management teams.
- Created executive-facing data visualizations delivering real-time manufacturing KPIs across production throughput, quality, and delivery performance.

Multi-Level Production Order & Subcontracting Architecture | *Business Central, AL*

- Designed end-to-end production order architecture handling cascading BOM splits (up to five levels), subcontracted plating operations, and scrap cost isolation, enabling accurate unit costing and manufacturing variance tracking.

Color Detector for Manufacturing Quality Control | *MATLAB, Image Processing*

- Built an image-processing tool to identify color-based manufacturing defects, supporting automated visual quality inspection on the production line.

EDUCATION

Bachelor of Engineering – Management Engineering (ABET-Accredited) 2017 – 2022

Istanbul Technical University (ITU) | QS World Top 350 | Turkey's #1 Engineering University (est. 1773)

Curriculum: Production Planning & Control, Supply Chain Management, Cost Accounting, Strategic Management, Quantitative Research & Data Analysis, Statistics, Python, MATLAB, Project Design & Management

Exchange Program: *Universitat Politècnica de Catalunya (UPC), Barcelona, Spain* | QS Top 400

CERTIFICATIONS & CREDENTIALS

Six Sigma Green Belt – The Council of Six Sigma
Spanish Certification (Advanced / C1) – Instituto Cervantes
WES Credential Evaluation – U.S. equivalency confirmed (Bachelor of Engineering)